

EXERCISES AND EXACT SOLUTIONS

Trust the Health Indicator Only After Testing Its Denominator

Independent practice, facilitated discussion, and assessment reference

Production Unit	PU-B07-C26
Notebook	V7_B_N06
Version	0.1
Assessment boundary	Learning evidence—not operational authorization

Instructions

Complete Part A before consulting the solutions. State assumptions explicitly. Where evidence is insufficient, write an abstention or escalation response rather than inventing precision.

Exercise 1

Detect duplicates.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exercise 2

Reconcile changing catchments.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exercise 3

Explain why missing is not zero.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exact solutions and scoring guide

Solution 1

Use event/report keys, timestamps, source identifiers, and duplicate-resolution rules.

Solution 2

Version facility catchments and population denominators by period.

Solution 3

Zero asserts no events; missing means no usable observation and requires follow-up or principled estimation with uncertainty.

Scoring rubric

Criterion	Strong evidence	Revision required
Decision framing	Owner, action, horizon, authority named	Model output substituted for decision
Evidence	Concepts, grain, provenance, limits explicit	Data treated as self-explanatory
Uncertainty	Abstention/escalation used appropriately	False precision or hidden limitations
Governance	Human authority and safeguards retained	Authority silently delegated